An Aerial-Ground Decision Support Framework for Salinity-Aware Sluice Gate Operation using Simulated UAV-Derived Spatial Monitoring

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Abstract. Salinity intrusion poses significant challenges to water management in coastal and deltaic regions, particularly in the Mekong Delta, where sluice-gate operations play a critical role in regulating freshwater availability. Existing studies primarily focus on salinity mapping and prediction, while the integration of spatial monitoring data into operational decision-making remains limited. This study proposes an aerial-ground decision support framework for salinity-aware sluice gate operation, integrating ground salinity observations, a simulated UAV-proxy spatial salinity layer, and a risk-aware rule-based decision model within a GIS environment. In this prototype, the UAV-proxy layer is generated using inverse distance weighting interpolation to emulate spatial salinity information that could be obtained from future UAV-based monitoring. A prototype system is implemented in QGIS using a four-layer architecture that includes data processing, spatial integration, decision support, and visualization. An experimental evaluation of approximately 2,100 decision instances demonstrates that incorporating UAV-derived spatial information improves operator agreement from 68.2% to 79.5%, reduces unsafe intake decisions (violation rate from 12.4% to 6.7%), and enhances decision stability. The results show that even simulated aerial salinity information can significantly improve decision quality compared with ground-only approaches.

Keywords: UAV-based monitoring; aerial-ground systems; decision support system; salinity intrusion; GIS; sluice gate operation; risk-aware modeling

1 Introduction

Salinity intrusion is a major challenge in coastal and deltaic regions, especially in the Mekong Delta, where sluice-gate operations are essential for regulating freshwater availability. The delta is highly vulnerable due to its low elevation,

dense river network, tidal influence, and reliance on rice-based agriculture. Increasing salinity caused by sea-level rise, reduced upstream flow, and climatic variability threaten agricultural productivity, freshwater supply, and land-use stability [1,2,3,4].

Recent studies have used hydrological modeling, remote sensing, machine learning, UAV imagery, and satellite data to map or predict salinity patterns [5,6,7,8]. These approaches provide useful spatial information but primarily focus on estimation, mapping, or prediction. In practice, salinity monitoring in the Mekong Delta remains weakly connected to gate-level operational decisions, which often rely on manual measurements and local communication. Spatial and temporal salinity variability further makes it difficult to generalize fixed decision rules across sluice gates.

To address this gap, we propose a GIS-based decision-support framework for salinity-aware sluice-gate operation. The framework integrates ground observations, sluice-gate spatial data, and a simulated UAV-proxy spatial salinity layer in QGIS. The UAV-proxy layer is generated from ground observations through spatial interpolation and is used to evaluate the decision-support value of spatial salinity context rather than to validate a physical UAV sensing platform.

The main contributions are: (i) an aerial–ground decision support framework for integrating spatial salinity information into sluice gate operation; (ii) a risk-aware rule-based model incorporating spatial context, temporal variation, and uncertainty; (iii) a lightweight QGIS prototype for interpretable, location-specific recommendations; and (iv) an empirical evaluation showing improved operator agreement, reduced unsafe decisions, and increased decision stability.

2 Literature Review

Salinity intrusion affects coastal agriculture by reducing water availability, plant growth, and crop yield [1]. In the Mekong Delta, salinity is intensified by sea-level rise, changes in upstream flows, and climate variability, affecting agricultural profitability and land-use decisions [2]. Remote sensing and machine learning have therefore been widely used for salinity mapping and prediction [5,9,10]. UAV and satellite observations can complement sparse ground measurements by providing spatially continuous information. Zhang and Zhao [6] integrated satellite, UAV, and ground measurements for coastal saline soil inversion; Xie et al. [7] combined UAV and satellite data for large-scale salinity monitoring; and Jiang et al. [8] used UAV hyperspectral imagery, Sentinel-2 imagery, and machine learning to estimate estuarine salinity. However, these studies primarily address salinity inversion, mapping, or prediction rather than operational sluice-gate decision support.

GIS-based analysis is useful for salinity management because salinity varies spatially and temporally. QGIS provides open-source capabilities for spatial analysis and visualization [11], while GIS-based systems have been applied to environmental monitoring and infrastructure analysis [12]. Spatial

decision support systems have also supported water-infrastructure planning, such as GIS-based prioritization of water-network interventions [13] and rural water-infrastructure planning [14]. However, these systems primarily support planning and assessment rather than dynamic sluice-gate operation.

Operational decision support also requires interpretability and alignment with expert judgment. Human-in-the-loop DSS studies emphasize expert oversight and agreement between system outputs and human decisions [15,16]. In addition, spatial estimates from sparse observations may contain interpolation and measurement uncertainty, motivating uncertainty-aware decision logic [17]. For future UAV deployments, studies on aerial-assisted edge computing highlight the importance of latency, bandwidth, energy, and edge processing [18,19]. This study addresses the remaining gap by linking ground observations, simulated UAV-proxy spatial salinity information, uncertainty-aware risk scoring, and GIS-based visualization to gate-level operational recommendations.

3 System Design and Methodology

3.1 System Architecture Overview

The proposed system is an aerial-ground GIS-based DSS designed to support salinity-aware sluice gate operation. It integrates ground salinity observations, simulated UAV-derived spatial salinity information, sluice gate spatial data, and rule-based decision logic within a QGIS environment. As shown in Fig. 1, the architecture comprises four components: the Data Layer, Processing Layer, Decision Support Layer, and Visualization Layer. Raw salinity observations are validated and spatially linked to sluice gate locations, evaluated by a rule-based engine, and finally visualized as operational status and recommendations in QGIS.

Data Layer. The data layer contains ground salinity observations $S(x^*, t)$, water conditions $W(x^*, t)$, sluice gate spatial data, simulated UAV-proxy salinity estimates $A(x^*)$, seasonal information $T(t)$, and location-specific rule sets $R_{loc}(x^*)$. Salinity and water-condition records are stored as CSV files, while sluice gates are represented as GIS point features with attributes such as gate name, coordinates, safe and critical salinity thresholds, and operational priority. Local rule sets are linked to gate features through gate identifiers or coordinates, enabling consistent integration of observations, spatial locations, and operational rules.

Processing Layer. The processing layer prepares input data for spatial analysis and decision evaluation using Python scripts and QGIS processing tools. Raw salinity records are validated based on numeric salinity values, timestamps, gate identifiers, and coordinates. Records with missing salinity values are marked as "No Data", while invalid identifiers or coordinates are excluded.

After validation, salinity observations are spatially joined with the sluice gate layer. To emulate aerial monitoring, the system generates a simulated UAV-proxy salinity raster using inverse distance weighting (IDW) interpolation from ground

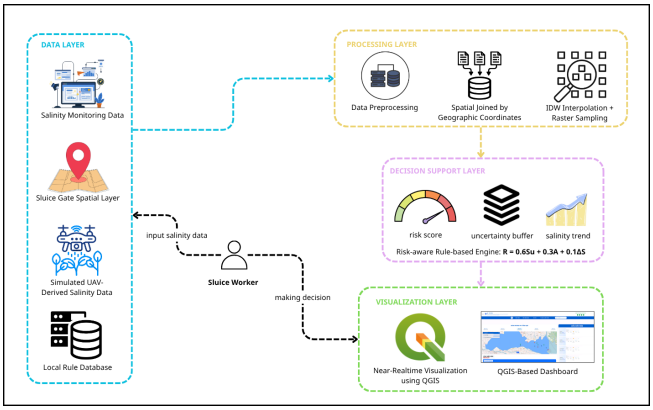


Fig. 1: Architecture of the proposed GIS-based decision support framework. In this prototype, the UAV-proxy spatial salinity layer is simulated using IDW interpolation from ground observations to evaluate the decision-support value of spatial salinity context.

observations. Since no physical UAV imagery is used, this raster is interpreted as a simulated spatial proxy rather than a directly observed aerial salinity product. To reduce circularity, the estimate for each target gate is generated from observations of the remaining gates, excluding the target gate's own salinity value. Raster values are then sampled at gate locations to obtain the simulated aerial salinity estimate $A(x^*)$.

The resulting spatio-temporal table contains gate location, ground salinity, simulated UAV-proxy salinity, timestamp, and water-condition attributes. Each sluice gate is treated as a discrete control point, preserving interpretability and matching real-world gate operation.

Decision Support Layer. The decision support layer implements a risk-aware rule-based engine that converts monitoring inputs into gate-level operational recommendations. Instead of relying only on fixed salinity thresholds, the proposed engine computes a continuous salinity risk score by combining ground salinity, simulated UAV-derived salinity, temporal salinity variation, and observation uncertainty.

For each sluice gate x^*, the normalized ground salinity value is denoted as $S_n(x^*, t)$ and the normalized simulated UAV-derived salinity estimate is denoted as $A_n(x^*)$. To reduce the effect of sensor noise, an uncertainty-adjusted salinity value is defined as

$$S_u(x^*, t) = S_n(x^*, t) + \lambda\sigma(x^*),$$

where $\sigma(x^*)$ represents the local observation uncertainty and λ is a conservative safety coefficient.

The salinity trend is computed as

$$\Delta S(x^*, t) = S_n(x^*, t) - S_n(x^*, t-1),$$

where a positive value indicates increasing salinity risk. The final risk score is then defined as

$$R(x^*, t) = 0.6S_u(x^*, t) + 0.3A_n(x^*) + 0.1\Delta S(x^*, t),$$

where the weights prioritize direct ground measurements while still incorporating aerial spatial context and temporal variation.

The operational decision is generated from the risk score

$$Decision = \begin{cases} \texttt{Water intake}, & R(x^*, t) < 0.4 \\ \texttt{Limited intake}, & 0.4 \leq R(x^*, t) < 0.7 \\ \texttt{Gate closure}, & R(x^*, t) \geq 0.7. \end{cases}$$

This design preserves the interpretability of rule-based decision-making while improving methodological rigor by accounting for spatial context, measurement uncertainty, and short-term salinity dynamics.

Visualization Layer. The visualization layer is implemented in QGIS. Each sluice gate is displayed as a point feature with updated attributes, including salinity level, timestamp, risk level, and recommended action. Categorized symbology is used to represent safe, warning, and critical conditions, allowing operators to identify locations where water intake is possible, should be restricted, or should be stopped. The attribute table also provides gate-level details for inspection and operational coordination.

3.2 System Implementation and Outputs

The system is implemented as a lightweight QGIS-based prototype with a task-oriented interface. Similar to the QGIS Light concept, unnecessary GIS functions are avoided to reduce user complexity [20]. The interface displays ground salinity, simulated UAV-proxy salinity, risk level, and recommended gate actions. The main outputs include spatial salinity visualization, categorized gate status, and location-specific recommendations for water intake, restriction, or closure. In a future physical UAV deployment, aerial observations would be transmitted from the UAV platform to a ground station or edge device, where preprocessing, georeferencing, and raster generation could be performed before updating the QGIS dashboard. Rather than transmitting raw imagery directly to the operator interface, the system could send processed salinity layers or sampled gate-level estimates to the decision-support module, reducing bandwidth demand and dashboard latency.

4 Experiment

4.1 Data Description

The dataset consists of approximately 52,000 hourly salinity observations collected at nine sluice gates over an eight-month period (January–August

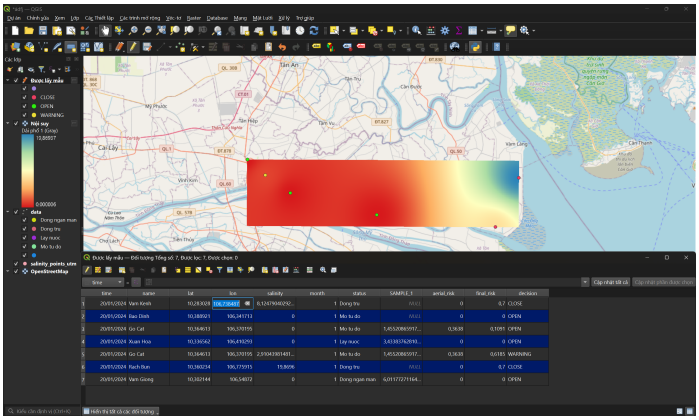


Fig. 2: QGIS prototype interface showing sluice gate attributes, salinity status, UAV-derived salinity estimate, and operational recommendation.

2024). Each record contains salinity $S(x^*, t)$, water level $W(x^*, t)$, location $x^* = (lat, lon)$, and timestamp t.

Because the raw dataset contains high-frequency observations, the data is aggregated at a daily level to reduce redundancy while preserving operational relevance. Each sluice gate, therefore, generates one representative decision per day based on daily salinity conditions, resulting in approximately 2,100 decision instances.

The evaluation further emphasizes boundary and high-risk cases in which salinity values are close to or exceed operational thresholds. These cases are operationally important because decision ambiguity is highest and additional spatial context is most useful.

A simulated UAV-derived salinity layer is generated using inverse distance weighting (IDW) interpolation from ground salinity observations. The resulting raster approximates a spatially continuous salinity surface and is sampled at each sluice gate to obtain an aerial salinity estimate $A(x^*)$.

Although the UAV-derived salinity data is simulated, it is designed to approximate realistic spatial monitoring scenarios and serves as a controlled setting to isolate and evaluate the impact of aerial spatial information on decision-making.

This enables comparison between ground-only and aerial-ground decision configurations.

4.2 Evaluation Design and Metrics

The evaluation examines whether UAV-derived spatial information improves decision quality beyond point-based threshold rules. The study does not report rule-matching accuracy, since comparing system outputs with the same

predefined rules would only verify software execution rather than decision usefulness.

Two configurations are compared:

- **Ground-only DSS**: decisions based on point salinity $S(x^*, t)$.
- **Aerial-ground DSS**: decisions based on both $S(x^*, t)$ and UAV-derived salinity $A(x^*)$.

Operator judgment is used as an operational reference rather than as an absolute ground truth. In this study, operator judgment is derived from historical gate-operation practice and local threshold interpretation provided by experienced sluice workers under similar salinity and water-level conditions. For each daily decision instance, the system recommendation is compared with the corresponding operator-preferred action. Therefore, the agreement metric should be interpreted as operator alignment rather than objective classification accuracy. This allows the evaluation to assess whether the DSS produces recommendations consistent with practical field experience, rather than merely verifying that the implemented rules are executed correctly. In future deployments, the reference standard should be strengthened using independent annotations from multiple experienced operators, inter-rater agreement analysis, and validation against historical gate-operation records.

The evaluation uses five metrics:

(1) Decision Distribution. This metric compares the proportion of `Water Intake`, `Limited Intake`, and `Gate Closure` decisions.

(2) Operator Agreement Rate.

$$Agreement = \frac{N_{matched}}{N_{total}} \times 100\%,$$

where $N_{matched}$ is the number of decisions consistent with operator judgment.

(3) Decision Change Rate.

$$Change\ Rate = \frac{N_{changed}}{N_{total}} \times 100\%,$$

where $N_{changed}$ is the number of decisions modified after incorporating UAV-derived salinity information.

(4) Threshold Violation Rate. This metric measures potentially unsafe intake recommendations. A violation occurs when the system recommends Water Intake while the final risk score exceeds the warning threshold or when salinity exceeds the local safe threshold.

$$Violation\ Rate = \frac{N_{unsafe}}{N_{total}} \times 100\%,$$

where N_{unsafe} denotes the number of intake recommendations made under elevated salinity risk.

(5) Decision Switching Rate. This metric measures how frequently recommendations change across consecutive decision periods. A lower switching rate indicates more stable decisions and lower sensitivity to short-term fluctuations.

4.3 Results and Discussion

Decision Distribution and Operator Agreement. As shown in Table 1, the aerial-ground DSS improves agreement with operator judgement from 68.2% to 79.5%. This suggests that incorporating UAV-derived spatial salinity information produces recommendations that better align with practical operating experience.

Table 1: Comparison of Ground-only and Aerial-ground DSS

Metric	**Ground-only DSS**	**Aerial-ground DSS**
Total Decisions	2,160	2,160
`Water Intake` (%)	38.5	35.2
`Limited Intake` (%)	34.1	36.8
`Gate Closure` (%)	27.4	28.0
Operator Agreement (%)	68.2	79.5
Decision Change Rate (%)	–	21.3
Threshold Violation Rate (%)	12.4	6.7
Decision Switching Rate	0.31	0.22

The decision distribution also changes moderately. Compared with the ground-only configuration, the aerial-ground DSS slightly reduces `Water Intake` decisions and increases `Limited Intake` decisions. This indicates a more cautious strategy in uncertain salinity conditions, where point-based measurements alone may underestimate surrounding salinity risk.

Decision Change and Risk Reduction. The aerial-ground DSS modifies 21.3% of decisions compared to the ground-only configuration. These changes are concentrated in boundary or spatially heterogeneous cases, where local salinity measurements do not fully represent surrounding conditions.

The threshold violation rate decreases from 12.4% to 6.7%, indicating fewer unsafe intake recommendations under high-risk salinity conditions. This result suggests that the UAV-derived salinity layer provides useful spatial evidence for refining gate-level decisions rather than simply duplicating point-based measurements.

Decision Stability. The decision switching rate decreases from 0.31 to 0.22 after incorporating UAV-derived salinity information. This indicates that the aerial-ground DSS produces more stable recommendations and is less sensitive to short-term local fluctuations. Such stability is important in sluice gate operation, where frequent changes in gate status may be impractical or operationally undesirable.

Sensitivity Analysis of Risk Weights. To examine whether the results depend strongly on the selected risk-weight configuration, a simple sensitivity analysis is conducted by varying the weight assigned to the UAV-derived salinity component. The baseline configuration uses $0.6S_u + 0.3A_n + 0.1\Delta S$.

Two alternative settings are tested by varying the UAV weight while keeping the ground observation as the dominant component.

Table 2: Sensitivity analysis of UAV-derived salinity weight

Risk weight setting	**Operator agreement**	**Violation rate**	**Switching rate**
$0.7S_u + 0.2A_n + 0.1\Delta S$	76.8%	7.4%	0.24
$0.6S_u + 0.3A_n + 0.1\Delta S$	79.5%	6.7%	0.22
$0.5S_u + 0.4A_n + 0.1\Delta S$	78.6%	6.9%	0.23

As shown in Table 2, the aerial-ground configuration improves operator agreement from 68.2% to 79.5%, reduces the threshold violation rate from 12.4% to 6.7%, and reduces the decision switching rate from 0.31 to 0.22. The decision distribution also becomes slightly more conservative, with fewer `Water Intake` recommendations and more `Limited Intake` recommendations. This suggests that the spatial salinity context helps reduce unsafe intake decisions, especially under boundary or heterogeneous salinity conditions.

Table 3: Ablation analysis of decision-support components

Configuration	**Operator Agreement**	**Violation Rate**	**Switching Rate**
Ground-only threshold	68.2%	12.4%	0.31
Ground + temporal trend	71.6%	10.8%	0.28
Ground + uncertainty adjustment	73.4%	9.1%	0.27
Ground + simulated spatial layer	76.9%	7.6%	0.24
Full risk-aware model	79.5%	6.7%	0.22

Table 3 shows that the results remain stable under moderate changes in the salinity weight of the UAV-proxy. The baseline setting achieves the best operator agreement and the lowest violation rate, while nearby configurations show only small differences in performance. This suggests that the improvement is not caused by a single arbitrary weight setting.

System Response Time. The system response time is approximately 5–8 seconds, including data aggregation, interpolation, raster sampling, decision computation, and QGIS layer refresh. The delay is mainly caused by QGIS rendering rather than decision computation. For deployment, latency can be reduced through incremental raster updates, cached map layers, asynchronous rendering, and edge preprocessing before importing UAV-derived spatial data into QGIS.

Visualization Output. Figure 3 illustrates representative dashboard outputs under safe, warning, and critical salinity conditions. Color-coded visualization supports rapid interpretation and comparison across multiple sluice gates.

For practical deployment, this latency can be reduced through incremental raster updates, cached map layers, asynchronous QGIS rendering, and separation of decision computation from visualization refresh. In addition, future UAV-derived spatial data could be preprocessed on an edge device or ground station before being imported into the QGIS dashboard. These optimizations would allow the decision engine to update gate recommendations without requiring full map re-rendering after every new observation.



(a) Safe condition (b) Warning condition (c) Critical condition

Fig. 3: Representative QGIS decision outputs under safe, warning, and critical salinity conditions. The displayed recommendations are generated from the risk-aware rule model using ground observations and simulated UAV-proxy spatial salinity information.

Overall, the results suggest that integrating simulated UAV-proxy spatial salinity information improves operator agreement, reduces unsafe intake recommendations, and increases decision stability compared with ground-only threshold rules. Although the UAV-proxy component is simulated, the experiment demonstrates how spatial salinity context can support more informed sluice gate operation. Future work will incorporate real UAV observations, predictive salinity models, and validation against historical operator decisions.

5 Conclusion

This study proposed an aerial-ground GIS-based decision support framework for salinity-aware sluice gate operation in the Mekong Delta. The system integrates ground salinity observations, a simulated UAV proxy spatial salinity layer, and an interpretable risk-aware decision model in QGIS.

Experimental results suggest that simulated spatial salinity information improves operator alignment and decision stability. Compared with the ground-only configuration, the aerial-ground configuration improves agreement with operator judgment from 68.2% to 79.5%, reduces unsafe intake decisions from 12.4% to 6.7%, and decreases the decision switching rate from 0.31 to 0.22. These results indicate that spatial salinity context is useful under boundary or heterogeneous salinity conditions where point observations alone may be insufficient.

The study has several limitations. The UAV-proxy salinity layer is simulated using IDW interpolation rather than obtained from physical UAV sensing, so the results should be interpreted as prototype-level evidence rather than validation of a deployed UAV salinity-monitoring system. Although leave-one-gate-out interpolation is used to reduce circularity, the simulated layer may not fully capture the noise, resolution, georeferencing error, and communication constraints of real UAV observations. Operator judgment is also used as an operational reference rather than an absolute ground truth, and the evaluation is limited to nine sluice gates and an eight-month observation period.

Future work will integrate real UAV observations, calibrate risk-weight parameters using larger expert-validated datasets, compare the rule-based

model with lightweight data-driven baselines, and optimize aerial-to-ground data transmission for near-real-time deployment. Overall, this work bridges spatial salinity monitoring and operational water management by linking simulated spatial salinity information, risk-aware decision logic, and GIS-based visualization into an interpretable sluice gate decision-support framework.

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